
Project Report- MAD-1

Student Details

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- Problem Statement: Quiz Master - V1

Project Details:

Project Description:

The Quiz Master application is designed to provide an interactive and structured platform for conducting quizzes. It enables user authentication, quiz creation, subject and chapter management, and scoring mechanisms. The primary goal is to create an engaging and user-friendly environment for both quiz administrators and participants.

Key Features:

- **User Authentication:** Secure login and registration for both administrators and quiz participants.
- **Profile Management:** Users can manage their profiles, including updating personal details.
- **Subject and Chapter Management:** Administrators can define subjects and chapters for better organization of quiz content.
- **Quiz Creation:** Admins can create quizzes associated with specific chapters, define quiz duration, and manage questions.
- **Scoring System:** Automatically calculates and stores quiz scores for user performance tracking.
- **Admin Dashboard:** Allows administrators to manage users, quizzes, and track overall system performance.

Approach:

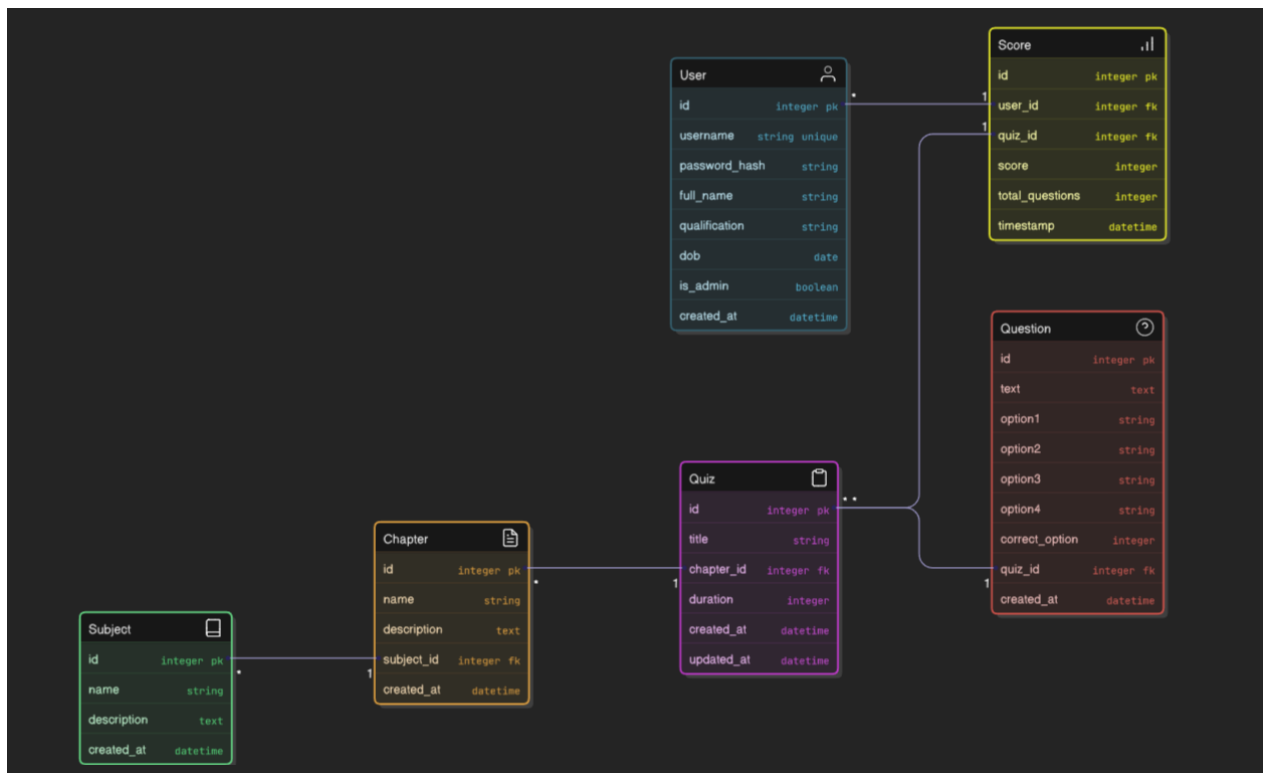
- **Requirement Analysis:** Identified essential features such as authentication, quiz and subject management, scoring, and reporting.
- **Design:** Developed a structured database schema and user-friendly UI to accommodate the application's core functionalities.

- **Implementation:** Utilized Flask for backend operations, SQLite for data storage, and Jinja2 with Bootstrap for front-end design.
- **Testing:** Conducted extensive unit and integration testing to validate system performance and user experience.

Frameworks and Libraries Used:

- **Flask:** Backend framework for handling application logic and routing.
- **Jinja2:** Templating engine for rendering dynamic HTML pages.
- **Bootstrap:** Front-end framework for responsive design and styling.
- **SQLite:** Lightweight database for storing quiz-related data.

ER Diagram of the Database:



API Resource Endpoints:

- **/users:** Retrieves all registered users.
- **/subjects:** Retrieves all subjects.
- **/chapters:** Retrieves all chapters under specific subjects.
- **/quizzes:** Retrieves all available quizzes.
- **/questions:** Retrieves all questions associated with a quiz.
- **/scores:** Retrieves scores of users from completed quizzes.

Drive Link of the Presentation Video:

For the presentation video, [click here](#).